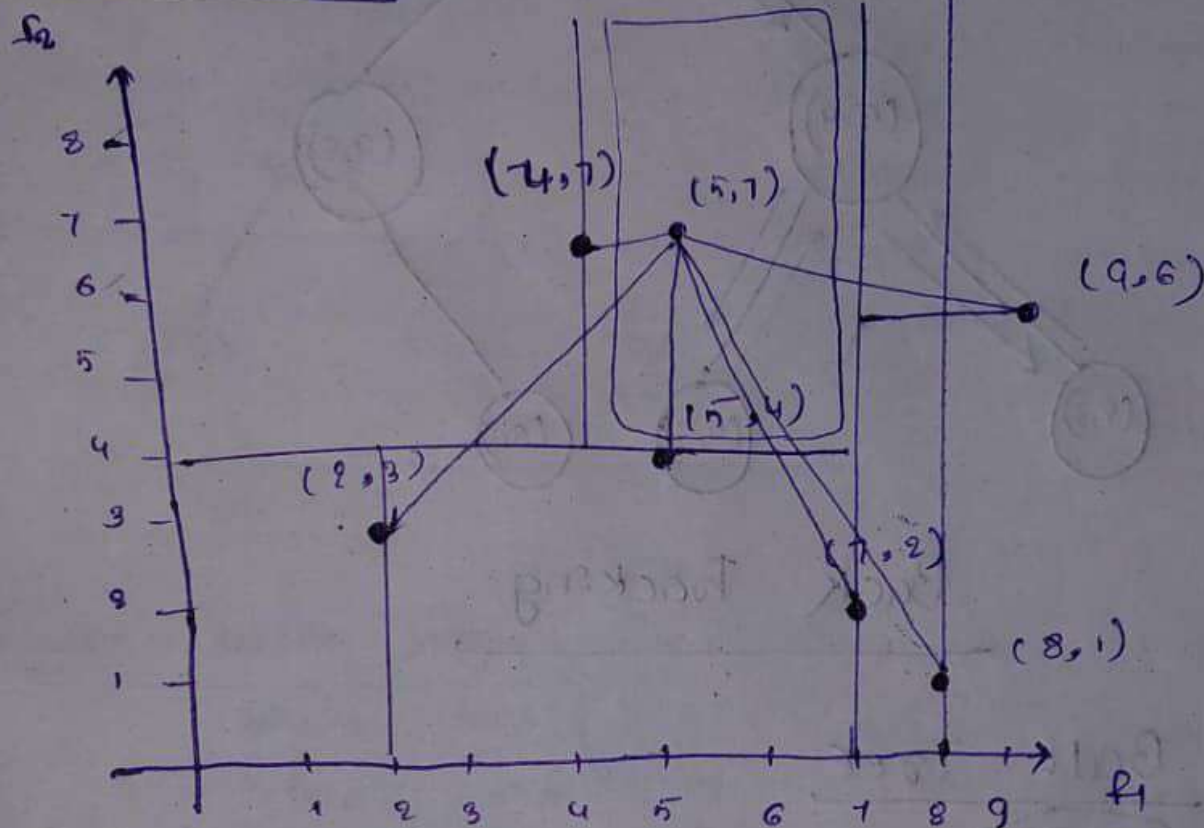


Variants of KNN

80

① KD Tree



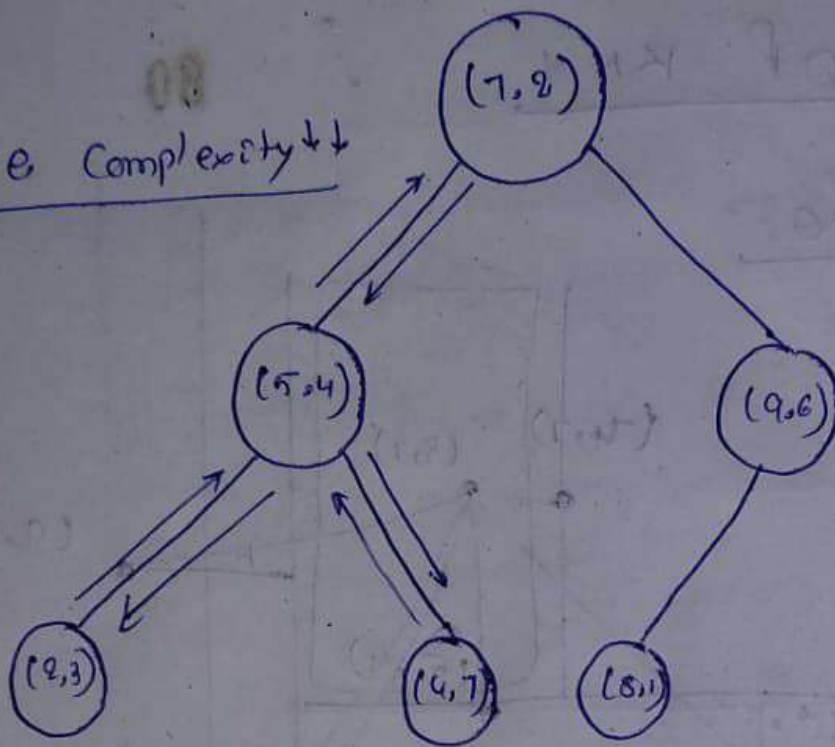
f_1	f_2
1	2
5	4
9	6
2	3
4	7
8	1
<u>Median</u>	<u>Median</u>

$$\begin{aligned} &\underline{f_1} \\ &2, 4, \boxed{5, 7}, 8, 9 \\ &\frac{5+7}{2} = \frac{12}{2} = 6 \end{aligned}$$

$$\begin{aligned} &\underline{f_2} \\ &1, 2, \boxed{3, 4}, 6, 7 \\ &\frac{3+4}{2} = \frac{7}{2} = 3.5 \end{aligned}$$

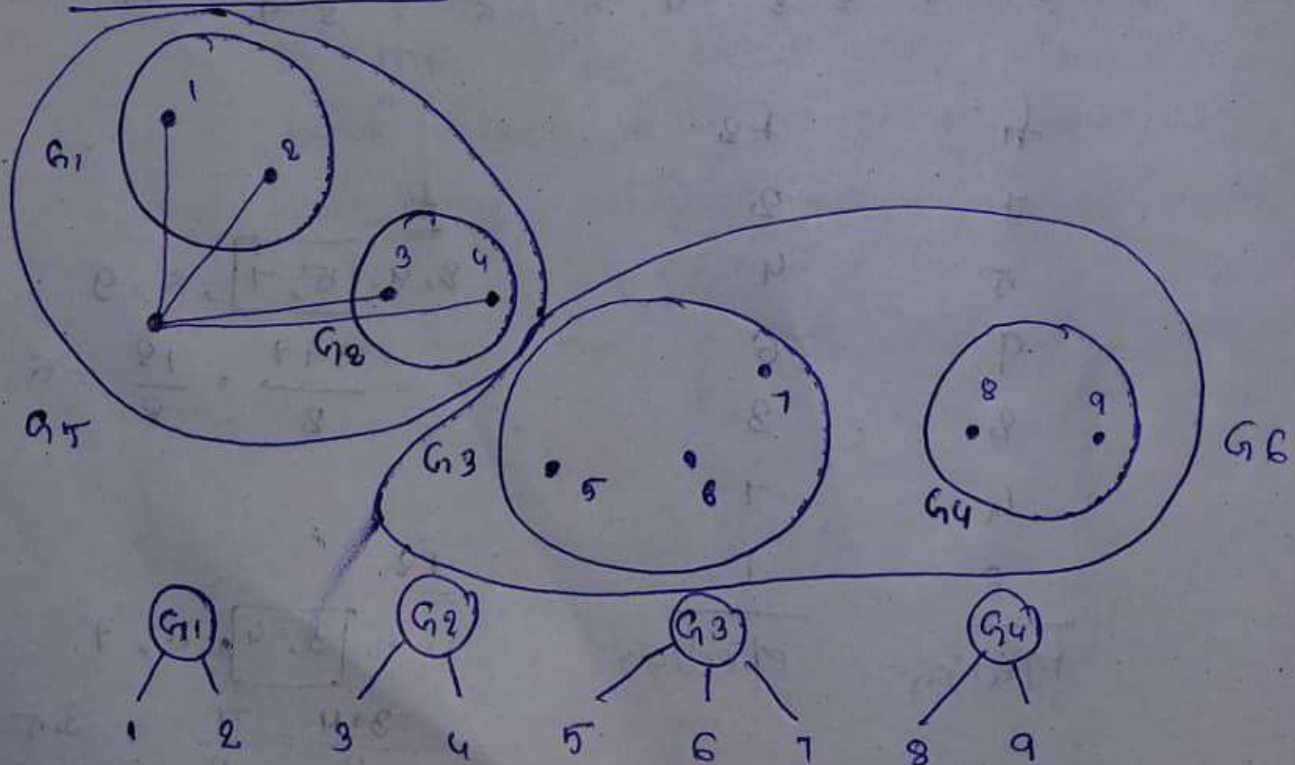
Binary Tree [KD Tree]

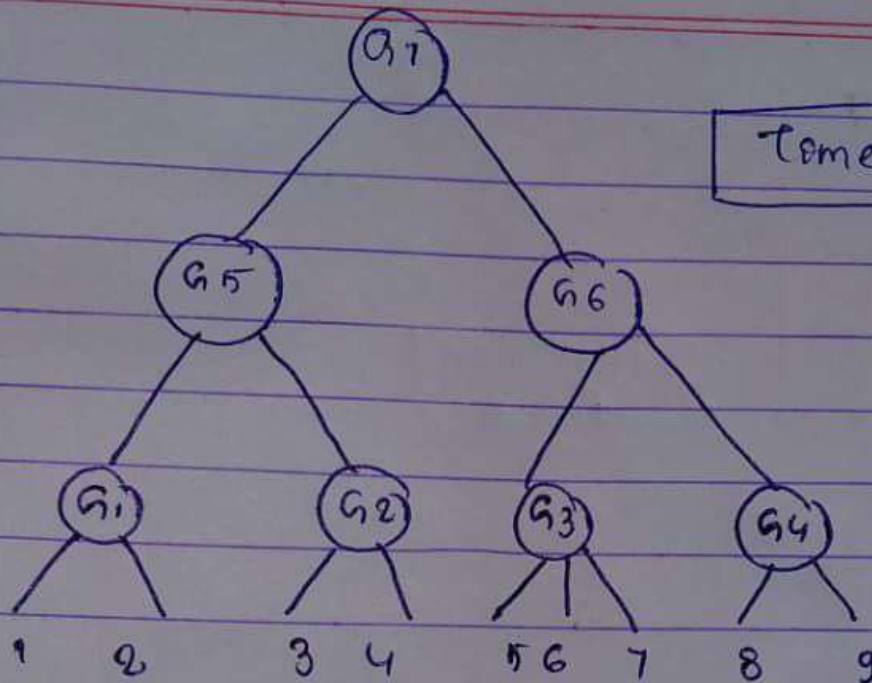
Time Complexity ↓↓



Back Tracking

② Ball Tree





Time Complexity $\downarrow\downarrow$

Notes: With respect to Binary tree the time complexity to ~~find every~~ search the element or find the distance (first of all search then calculate the distance) is reduces because here we no need to calculate the distance with each on every point.